

Learning Interpretable Disease-relevant Gene Programs for Predicting Personalized Disease Risk

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Prediction of personalized disease risk can inform interventions that delay onset or reduce severity in individual patients. Gene expression profiles capture disease-relevant molecular processes that link genetic and environmental effects to clinical disease. However, existing expression-based risk prediction models either do not explicitly model the coordinated gene activity underlying cellular processes or yield predictions that are difficult to interpret. We introduce a supervised Bayesian Poisson factorization model for jointly learning disease-relevant gene programs (DRGPs) and predicting personalized disease risk from RNA-seq data. The DRGP model decomposes a gene expression matrix into latent factors representing gene programs that can be informed by prior pathway information, while an L1-penalized logistic regression encourages sparsity over DRGPs and links factor loadings and auxiliary covariates to disease labels. A variational Bayes algorithm is derived to enable efficient inference. We evaluate our method on simulated and experimental single-cell RNA-seq COVID-19 data and GTEx whole-blood bulk RNA-seq data that include genetic risk covariates. The DRGP method achieves competitive predictive performance while recovering biologically coherent gene programs aligned with validated biomarkers, suggesting that accurate personalized disease risk prediction is achievable without sacrificing interpretability. The source code for the DRGP method, simulation scripts, and appendix are freely available at <https://github.com/bayesomicslab/DRGP>.

Keywords: Personalized Disease Risk Prediction; Bayesian Hierarchical Modelling; Disease-Relevant Gene Programs; Supervised Poisson Factorization

1. Introduction

Predicting personalized disease risk is a central challenge in clinical decision-making, informing prevention, diagnosis, and treatment strategies for individual patients. Early prediction enables preventive interventions that can alter disease trajectories before irreversible pathology develops. For example, early detection and treatment of type 2 diabetes and its risk factors reduce cardiovascular complications and mortality relative to delayed diagnosis^{1,2}, and earlier

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initiation of statin therapy in high-risk individuals is associated with substantially lower rates of major cardiovascular events than delayed treatment³.

Traditional disease risk prediction (DRP) models, such as the Framingham risk score⁴ and the pooled cohort equations⁵, rely on clinical risk factors including age, sex, cholesterol, blood pressure, and comorbidities. More recently, polygenic risk scores (PRS) have emerged as a means to incorporate inherited genetic risk into DRP^{6,7}; these scores quantify genetic liability as a weighted sum of risk alleles previously associated with the disease. While PRS have some clinical utility⁸, the proportion of disease variance typically explained is substantially lower than estimated phenotypic variance from twin study estimates of heritability^{9,10}. Further, although the genome is largely fixed, disease risk can change over time in response to sociobehavioral factors or gene-by-environment interactions^{11,12}.

Since genetic and environmental risk factors can be mediated through gene expression changes^{13,14}, an increasingly common strategy for DRP is to augment traditional features with gene expression, which captures disease-relevant molecular processes not reflected in established risk scores^{15–17}. For example, transcriptomic signatures from circulating monocytes have been shown to improve prediction of atherosclerotic cardiovascular disease risk¹⁸. However, predicting disease risk from gene expression poses new methodological and biological challenges. First, gene expression data are high-dimensional and highly correlated due to the coordinated gene activity underlying cellular processes, whereas classical DRP uses a small number of weakly correlated clinical covariates^{4,5}. Second, because DRP models are intended for clinical use, interpretability of both model structure and predictions is essential, discouraging the use of complex models for which post-hoc explanations are not a reliable substitute for inherent interpretability¹⁹. Lastly, gene expression varies substantially due to factors unrelated to disease risk, including cell-type^{20,21}, environmental exposures^{22,23}, and immune activation^{24,25}, so only a small subset of genes is expected to be disease-relevant.

To address these challenges, expression-based DRP models typically pair feature-selection methods with linear predictors, reducing overfitting while maintaining model interpretability. Generalized linear models are widely adopted to estimate disease risk^{5,26,27}. Machine-learning models, such as boosting classifiers, have also been applied to DRP^{28,29} and can capture more complex relationships among features³⁰, although often at the expense of interpretability (see §A for additional details). Across these approaches, the simplifying assumptions of additive linear effects^{4,5} or unstructured interactions among gene expression features^{28,29} poorly align with the coordinated gene activity driving disease-relevant cellular processes.

Related but largely orthogonal unsupervised approaches aim to infer *gene programs*, sets of co-regulated genes representing cellular processes, from gene expression data. Matrix factorization methods such as hierarchical Poisson factorization³¹ model expression using latent factors, providing a structured representation of sparse count data that moves beyond the gene-independent assumptions common in expression-based DRP. Relative to individual-gene predictors, program-level representations reduce dimensionality, mitigate multicollinearity, and provide direct biological interpretability, since genes associated with a program are directly encoded in the latent factors and prior biological knowledge can be incorporated to guide factor learning³². However, these methods are unsupervised and unimodal: latent fac-

tors are optimized to reconstruct expression only, not to maximize disease discrimination. Downstream classifiers fit on unsupervised factors can achieve competitive prediction, but disease-irrelevant variation may dominate the latent space and reduce both predictive efficiency and interpretability.

In this work, we develop a supervised Bayesian Poisson factorization model that combines the interpretable representations of gene programs with optimization for disease relevance (Fig. 1). We define disease-relevant gene programs (DRGPs), which are subsets of genes that collectively contribute to some biological process relevant to disease onset or progression and are optionally informed by pathway information. We associate sample usage of DRGPs with disease using a regularized generalized linear model that allows for multi-modal interactions between DRGPs, traditional DRP covariates, and genetic variants. We derive a variational Bayes inference algorithm using Pólya-Gamma augmentation to restore conjugacy of the supervised term, to efficiently compute DRGPs. We benchmark DRP on simulated and experimental bulk and single-cell RNA sequencing (RNA-seq and scRNA-seq, respectively). Simulations combine a disease liability model with scDesign3³³ for single-cell and SPsimSeq³⁴ for bulk data. Experimental evaluations examine COVID-19 disease severity in scRNA-seq³⁵ and heart disease from GTEx whole blood data³⁶. Simulations show that our DRGP method recovers disease-relevant programs more accurately than unsupervised factorization when informative pathway priors are available, and that the inclusion of multi-modal information (e.g., genetic covariates) improves expression-based DRP performance. On experimental data, DRGP achieves predictive performance competitive with unsupervised-factor-plus-classifier and gene-level baselines while learning programs that are reproducible across random initializations and interpretable at the program level.

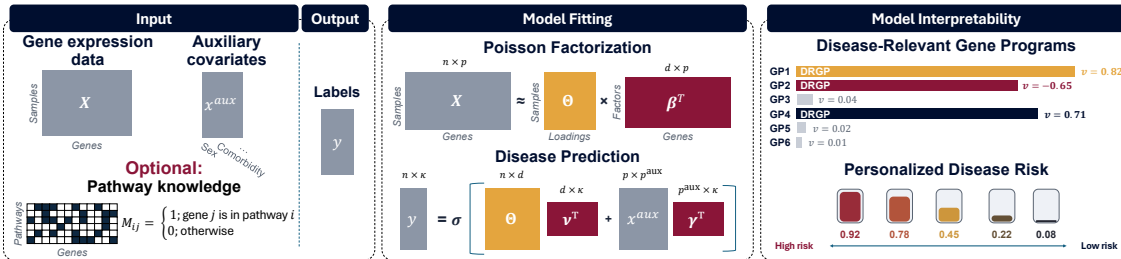


Fig. 1. **DRGP model overview.** From a gene expression count matrix $\mathbf{X}$, labels $\mathbf{y}$, covariates $\mathbf{x}^{\text{aux}}$, and an optional pathway mask $\mathbf{M}$ (left), the DRGP model simultaneously computes a Poisson factorization $\mathbf{X} \approx \Theta \beta^T$ and disease risk $\mathbf{y} = \sigma(\Theta \mathbf{v}^T + \mathbf{x}^{\text{aux}} \gamma^T)$ (center), yielding programs ranked by regression coefficient $|v|$ (large values indicate disease-relevance) and per-patient risk scores (right).

2. Methods

Assume the data contain m individuals, where individual i contributes n_i expression samples (Fig. 1, Left). In scRNA-seq, n_i represents the number of single-cell samples for individual i , whereas $n_i = 1$ for all i in bulk RNA-seq. Let $\mathbf{X}$ be an $n \times p$ gene expression matrix where $n = \sum_{i=1}^m n_i$ denotes the total number of samples, p is the number of genes, and row $\mathbf{x}_j$ and column $\mathbf{x}_{.g}$ denote sample j and gene g , respectively. Our model assumes there exists a d -dimensional

latent factor for each sample $\theta = \{\theta_1, \dots, \theta_j, \dots, \theta_n\}$ and each gene $\beta = \{\beta_1, \dots, \beta_g, \dots, \beta_p\}$. Sample- and gene-specific latent factors generate x_{jg} , *i.e.* the expression of gene g in sample j . Each individual is associated with a κ -dimensional vector $\mathbf{y}_i \in \{0,1\}^\kappa$ denoting the phenotype labels for individual i and a vector of observed auxiliary covariates, $\mathbf{x}_i^{\text{aux}}$.

2.1. The DRGP Model

The DRGP model generative process couples hierarchical Poisson factorization with logistic regression through the learned sample factor matrix (Fig. 1, Middle, Fig. 2, and Table 1).

For each sample j :

- Generate sample activity $\xi_j \sim \text{Gamma}(\alpha_\xi, \lambda_\xi)$.
- For each factor, generate sample factors $\theta_{j\ell} \sim \text{Gamma}(\alpha_\theta, \xi_j)$.

For each gene g :

- Generate gene inclusion rate $\pi_g \sim \text{Beta}(\alpha_\pi, \beta_\pi)$.
- Generate gene activity $\eta_g \sim \text{Gamma}(\alpha_\eta, \lambda_\eta)$.
- For each factor ℓ :
 - * Generate gene-program membership $m_{g\ell} \sim \text{Bernoulli}(\pi_g)$.
 - * Generate slab loading $\tilde{\beta}_{g\ell} \sim \text{Gamma}(\alpha_\beta, \eta_g)$.
 - * Set spike-and-slab prior $\beta_{g\ell} = m_{g\ell} \tilde{\beta}_{g\ell}$.

For each sample j and gene g :

- Generate gene expression $x_{jg} \sim \text{Poisson}(\theta_j \beta_g^\top)$.

For each phenotype k and factor ℓ :

- Generate latent factor regression coefficients $v_{k\ell} \sim \text{Laplace}(\mu_v, b)$.

For each phenotype k :

- Generate auxiliary regression coefficients $\gamma_k \sim \text{Normal}(\mu_\gamma, I\sigma_\gamma^2)$.

For each sample j and phenotype k , generate

$y_{jk} \sim \text{Bernoulli}\left(\sigma\left(\theta_j v_k^\top + x_j^{\text{aux}} \gamma_k^\top\right)\right)$, where σ is the logistic function.

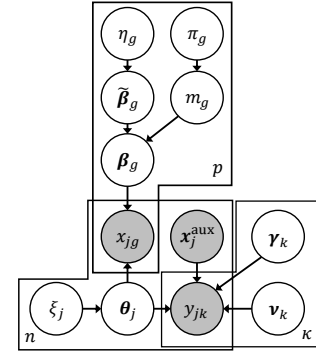


Fig. 2. **Supervised Poisson Factorization graphical model.** We omit the hyperparameters for brevity. We note that $\mathbf{X}^{\text{aux}}$ is an $n \times p^{\text{aux}}$ matrix of auxiliary covariates (e.g., age or sex).

2.2. Variational Inference Algorithm

Exact posterior inference is intractable due to the non-conjugacies introduced by the logistic supervision and sparsity inducing priors. We therefore derive a mean-field coordinate-ascent variational inference algorithm³⁷, restoring conjugacy of the supervised term exactly by Pólya-Gamma augmentation³⁸ (Algorithm 1). Training sequentially updates the gene-program parameters, sample-specific program activities, and supervised regression parameters. Prior path-

Table 1. Summary of model variables.

Param.	Dim.	Description
θ	$n \times d$	factor matrix for $\mathbf{X}$ and design matrix for $\mathbf{Y}$
β	$p \times d$	loading matrix for $\mathbf{X}$
$\mathbf{v}$	$\kappa \times d$	regression coefficients for θ
γ	$\kappa \times p^{\text{aux}}$	regression coefficients for x_{aux}
$\mathbf{y}$	$m \times \kappa$	individual phenotype
$\mathbf{x}$	$n \times p$	gene expression matrix
$\mathbf{X}^{\text{aux}}$	$m \times p^{\text{aux}}$	auxiliary factors (e.g., sex)

way knowledge enters through a binary mask $M \in \{0,1\}^{p \times d}$ with $M_{g\ell} = 1$ if gene g belongs to pathway ℓ ; entries with $M_{g\ell} = 0$ and excluded from all updates, giving three modes: *de novo* (no mask), pathway-guided (all d columns masked), and combined (d_{path} masked columns alongside $d - d_{\text{path}}$ free columns). Pólya–Gamma augmentation preserves conjugacy of the regression coefficients, while the phenotype contributes to each sample activity through a supervised correction that couples gene program activity and disease status (Eq. B.10).

For a held-out sample, all global parameters are fixed and only its local program activities $(\theta_{j^*}, \xi_{j^*})$ are inferred from expression. Disease risk is then computed from the posterior mean and variance of the logistic linear predictor. Each training sweep is dominated by the sparse count-allocation update and scales as $\mathcal{O}(\text{nnz}(\mathbf{X})d)$ where $\text{nnz}(\mathbf{X}) = |\{(j,g) : x_{jg} \neq 0\}|$. Memory scales as $\mathcal{O}(\text{nnz}(\mathbf{X}) + (n+p)d)$. The variational parameter update derivations, detailed inference and prediction algorithms, and complexity analysis are given in Supplement §B (Algs.2,3).

Algorithm 1 DRGP model training and test set prediction

Input: Expression $\mathbf{X}$, labels $\mathbf{Y}$, covariates $\mathbf{X}^{\text{aux}}$, number of programs d , and pathway mask $\mathbf{M}$

Output: Fitted variational parameters and test set disease risk estimates

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1: Initialize variational parameters for  $(\tilde{\beta}, \mathbf{m}_g, \pi, \eta, \theta, \xi, \mathbf{v}, \gamma)$  and enforce the pathway mask  $\mathbf{M}$ 
2: while the ELBO has not converged do ▷ Training loop
3:   Update variational parameters for  $(\tilde{\beta}, \mathbf{m}_g, \pi, \eta, \theta, \xi, \mathbf{v}, \gamma)$  using equations B.3 to B.14
4: end while
5: for each held-out sample  $j^*$  do ▷ Test set prediction
6:   while the local Poisson objective has not converged do
7:     Update  $(\theta_{j^*}, \xi_{j^*})$  from test set expression data  $\mathbf{x}_{j^*}$  using Equations B.8 and B.16
8:   end while
9:    $\hat{y}_{j^*k} \leftarrow \sigma \left( \frac{\mathbb{E}_q[A_{j^*k}]}{\sqrt{1 + \frac{\pi}{8} \text{Var}_q(A_{j^*k})}} \right)$  ▷ Equation B.17
10: end for
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2.3. Simulation Algorithm

We assess gene program recovery, DRP, and recovery of covariate effects on synthetic benchmarks built from experimental data. The design proceeds in four stages: (1) generating samples using scDesign3³³ on single-cell and SPsimSeq³⁴ on bulk data; (2) simulating nuisance and DRGPs; (3) perturbing sample gene expression; and (4) labeling samples as cases or controls based on DRGP burden. As the simulation procedure in step 1 depends on the dataset, we defer a full description to the results (Sec. 3).

Simulated Gene Programs. Given a simulated expression matrix X^0 , we simulate d^* gene programs $\{S_\ell\}_{\ell=1}^{d^*}$. Each program ℓ is defined by a support set of genes S_ℓ and a parameter $\phi_{g\ell} \in \mathbb{R}$ that determines the magnitude of expression changes induced when the program is active. Programs are partitioned into disease-relevant ($\ell \in \mathcal{D}$) and nuisance ($\ell \in \mathcal{N}$) programs. Both program types perturb gene expression through the same mechanism, but only DRGPs contribute to phenotype generation. This separation enables evaluation of whether methods can distinguish transcriptional variation associated with disease from unrelated biological vari-

ation. Program gene sets are either pathway-guided or *de novo*, with genes sampled uniformly without replacement. The gene support for pathway-guided programs, S_ℓ , is determined by a binary mask M defined by a known pathway (§2.2); *de novo* programs remain hidden from all methods and must be discovered directly from the simulated expression data.

Assigning Programs to Samples. Disease-associated transcriptional programs ($S_\ell : \ell \in \mathcal{D}$) are assigned to individuals rather than to cells. We draw each individual’s latent disease propensity as $p_i \sim \text{Beta}(\alpha_p, \beta_p)$ where larger values increase the prevalence of disease-relevant programs across that individual’s cells. To allow for program-specific variability across programs, we define $\tilde{p}_{i\ell} = \sigma(\sigma^{-1}(p_i) + u_{i\ell})$ where $u_{i\ell} \sim \mathcal{N}(0, \sigma_u^2)$. Disease-relevant programs are then assigned to individuals according to $c_{i\ell} \sim \text{Bernoulli}(q_0 + (q_1 - q_0)\lambda_\ell \tilde{p}_{i\ell})$, $\ell \in \mathcal{D}$, where $(0 \leq q_0 \leq q_1 \leq 1)$ are the baseline and maximum carrier probabilities, respectively, and $\lambda_\ell \in [0, 1]$ modifies program prevalence. As a result, individuals with larger disease propensities have a higher probability of carrying DRGPs. In contrast, nuisance programs are assigned independently of disease propensity, ensuring that nuisance programs affect gene expression but not disease status: $c_{i\ell} \sim \text{Bernoulli}(\pi_\ell)$ where $\ell \in \mathcal{N}$.

Gene Expression Perturbation. After program assignment, we perturb expression based on the data modality. Bulk RNA-seq activates programs at the individual level, whereas scRNA-seq simulations restrict activation to responder cell types within carrier individuals. Let μ_{jg}^0 denote the background expression mean for sample j and gene g . Active programs perturb the background expression on the log-mean scale according to

$$\log \mathbb{E}[x_{jg} \mid X^0] = \log \mu_{jg}^0 + \sum_{\ell} a_{j\ell} \phi_{g\ell}, \quad (1)$$

where $a_{j\ell}$ denotes the activation of program ℓ in sample j and $\phi_{g\ell}$ perturbs the expression of gene g in program S_ℓ additively; $\phi_{g\ell} = 0$ for genes outside the program support. We then resample counts from the perturbed means, restricting scRNA-seq to selected responder cell types and a fraction of their cells. The resampling procedure preserves the characteristics of the background expression data while incorporating the simulated transcriptional perturbations.

Label Generation. After expression perturbations, phenotype labels are generated from DRGP activity using a liability-threshold model. Program activations are first standardized across each individual’s samples, $Z_i = \sum_{\ell \in \mathcal{D}} \omega_\ell \tilde{a}_{i\ell}$ and $\tilde{Z}_i = \frac{Z_i - \bar{Z}}{\text{sd}(\tilde{Z})}$, where ω_ℓ denotes the contribution of DRGP ℓ to disease liability, and $\tilde{a}_{i\ell}$ represents the activation of program ℓ in individual i standardized across samples. We set $a_{i\ell} = a_{j\ell}$ for bulk data and aggregate sample program activations to produce an individual activation $a_{i\ell} = |C_i|^{-1} \sum_{j=1}^{n_i} a_{j\ell}$, where C_i is the set of cells of individual i . Disease liability is then defined as

$$l_g = \sqrt{w_Z} \tilde{Z}_i + \sqrt{w_s} \tilde{s}_i + \sqrt{1 - w_Z - w_s} \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1), \quad (2)$$

where $\tilde{s}_i$ denotes an optional standardized auxiliary covariate and w_Z and w_s specify the proportions of liability variance explained by transcriptomic programs and auxiliary covariates, respectively. Because $\tilde{Z}_i$, $\tilde{s}_i$, and ϵ_i are independent with unit variance, the square-root scaling makes w_Z , w_s , and $1 - w_Z - w_s$ their respective fractions of total liability variance. Disease labels are generated by thresholding the liability, $D_i = \mathbb{1}[l_i > \tau]$, where τ sets disease prevalence. Because experimental data lack cell-level labels, each cell inherits its individual’s label.

3. Results

We evaluated our DRGP method on both single-cell and bulk RNA-seq cohorts, each used for both synthetic and experimental evaluation. Synthetic benchmarks enable evaluation of program recovery and auxiliary-effect attribution because the true DRGPs are known. Experimental cohorts allow for disease-risk prediction and biological interpretation on real disease phenotypes. We compared three DRGP configurations (unmasked, masked, and combined) with two gene-program methods, scHPF³¹ and Spectra³², and two general matrix-factorization methods, NMF and PCA. The unsupervised baselines were followed by L_1 -regularized logistic regression. Where applicable, we also included gene-level L_1 -regularized logistic regression; model-selection details are provided in Appendix §D.

3.1. Datasets and Preprocessing

COVID-19 PBMC scRNA-seq. Single-cell experiments used the COVID-19 peripheral blood mononuclear cell (PBMC) atlas³⁵, comprising 846,847 cells from 148 COVID-19 patients spanning “mild or moderate” and “severe or critical” disease labels. Unless otherwise noted, analyses used unnormalized UMI counts. Gene symbols were mapped to Ensembl (v. 116)³⁹. We removed duplicate genes, genes not mapped to Ensembl, non-protein-coding genes, and genes detected in fewer than 0.1% of cells, yielding 13,770 genes. Disease severity was predicted at the individual level, with sex and three comorbidity indicators (asthma or chronic obstructive pulmonary disease, cardiovascular disease, and diabetes) included as auxiliary covariates.

GTEx Whole Blood. Bulk experiments used GTEx v8 whole-blood RNA-seq data³⁶. Because GTEx lacks a single cardiovascular phenotype, we defined a binary label equal to 1 for any history of prior myocardial infarction or acute coronary syndrome (MHHRTATT), ischemic heart disease or coronary artery disease (MHHRTDIS), or secondary heart-disease history indicator (MHHRTDISB). After removing patients with missing transcriptomic and phenotype information, the experimental cohort comprised 177 cases from a total of 708 individuals. Unless otherwise noted, we used raw read counts from protein-coding genes detected in at least 10% of samples, resulting in 27,693 genes. Auxiliary covariates included sex, four race indicators, standardized age and BMI, smoking status, hypertension, and type 2 diabetes.

3.2. Simulation Instantiation

Here, we describe modality-specific instantiations of the general simulation framework (§2.3).

COVID-19 Simulation. We generated realistic background expression by fitting scDesign3 to healthy-control PBMCs³³. Each simulated dataset consists of 80 individuals, 100 cells per individual across six cell types, and 10,000 genes per cell. Simulations include two annotated and two *de novo* DRGPs, plus one *de novo* nuisance program; each *de novo* program contains 50–100 genes. Individuals are assigned DRGPs, which are activated only within predefined responder cell types for a random fraction of $\rho = 0.3$ eligible cells. Program perturbations are added to the background expression (Eq. 1), and patient labels are generated using our simulation liability (§2.3) at a target prevalence of $\tau = 0.5$. We vary the perturbation strength

relative to cell-type variation $r = \delta^2 \frac{SS_{\text{pert}}}{SS_{\text{type}}} \in \{0.05, 0.15, 0.30\}$, where SS_{pert} is the variation induced by the simulated program, SS_{type} is the between cell-type variation, and δ is calibrated to achieve the target value of r . We parametrize the single-cell arm by r rather than a raw effect size because cell-type identity dominates single-cell expression variance, so a program’s detectability depends on its magnitude relative to that structure. Finally, we vary the transcriptomic contribution to disease liability $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$, generating five independent datasets for each combination of w_Z and r .

Bulk Simulation. Background expression is generated using SPsimSeq fit to GTEx whole blood samples³⁴, yielding 2,000 samples each with 5,000 genes. Each simulation uses six DRGPs, two annotated and four *de novo*, together with two nuisance programs, each supported on 50–150 genes. Programs are assigned to samples and directly perturb bulk expression. Samples include genetic covariates based on a coronary artery disease PRS (ID: PGS000018, meta-GRS_CAD)⁴⁰ computed from GTEx donor genotypes (aligned to GRCh38 using `plink2`⁴¹). For each GTEx donor with genetic data, we computed $PRS_i = \sum_v w_v g_{iv}$, where w_v is the PRS effect weight of variant v and genotype $g_{iv} \in \{0, 1, 2\}$ denotes the effect-allele dosage of individual i at variant v . The 866 donor PRS values were z-scored and upsampled to 2,000 synthetic expression samples by independently sampling with replacement from the PRS pool. Because the score affects disease liability but not expression (Eq. 2), w_s represents the fraction of liability variance explained by transcriptome-independent genetic risk.

Disease labels are generated using the liability model with both transcriptomic and auxiliary genetic contributions at a target prevalence of $\tau = 0.26$. We generate five simulation replicates for transcriptomic liability share $w_Z = 0.3$, and auxiliary liability share $w_s = 0.1$, and program effect size $\varphi \in \{\log_2 3, 3.0, 4.5\}$. Full parameter settings for both simulation arms are listed in Table D1.

3.3. Simulation Results

Recovery of Simulated Gene Programs. We evaluated DRGP recovery by matching inferred programs to ground truth programs using the Hungarian algorithm with cosine similarity. In our model, gene program score vectors are defined as the top 50 genes ranked by $\hat{s}_{g\ell} = \mathbb{E}_q[\beta_{g\ell}]\mathbb{E}_q[\eta_g]$. The area under the precision-recall curve (AUPRC) is then computed for the inferred programs matched to the true DRGPs (see Appendix C for more details).

In the scRNA-seq simulations, methods that leverage pathway annotations (DRGP-masked, DRGP-combined, and Spectra) recovered ground-truth gene programs more accurately than methods without them (DRGP-unmasked, scHPF, NMF, and PCA; Fig. 3A). Program recovery AUPRC was insensitive to varying w_Z and r except for a slight upward trend for Spectra (Fig. E1); methods performed similarly across varying responder fractions ρ (Fig. E2). All methods recovered *de novo* programs at near-chance levels, likely because their signals were obscured by pathway and cell-type variation inherited from the experimental scRNA-seq background. Recovery was generally higher in bulk data, where programs perturb expression directly at the subject level (Fig. 3B). Spectra and scHPF performed best for annotated programs, while PCA, DRGP-combined, scHPF, Spectra, and DRGP-unmasked were broadly comparable for *de novo* recovery.

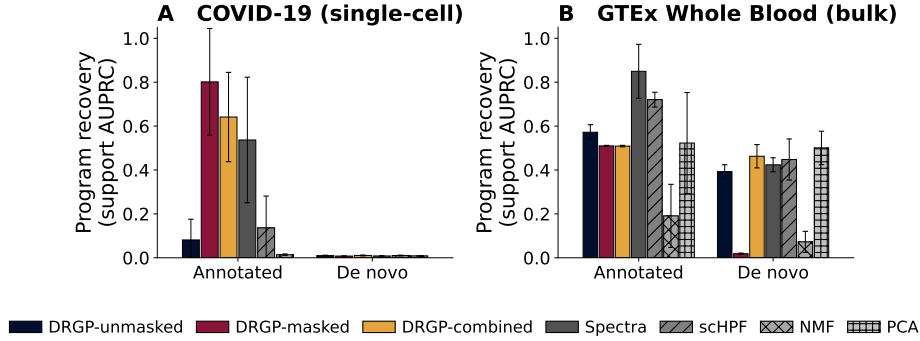


Fig. 3. **Recovery of simulated DRGPs across data modalities and prior-knowledge settings.** AUPRC is compared for (A) COVID-19 single-cell and (B) GTEx whole-blood simulations. Bars are means over five simulated draws and error bars show ± 1 SD.

Disease Prediction. We next evaluated DRP in the simulated data. In the scRNA-seq simulations, patient-level AUC increased with ρ , r , and w_Z (Figs. E3, E4), but cell-level performance remained near chance across methods, likely because cells inherited noisy patient-level phenotype labels (Fig. E5). We then investigated whether cell-level predictions could effectively be aggregated to form a single prediction for the individual, and found that methods allowing for both pathway and novel gene programs performed the best (DRGP-combined and Spectra; Fig. 4A). In the bulk simulations, our method remained competitive with the strongest factorization baselines (Fig. 4B).

Attribution of Multi-modal Auxiliary Effects in GTEx simulations.

Disease risk may be driven by both genetic factors and disease-associated transcriptional programs. An interpretable model should distinguish these sources of variation rather than allowing genetic effects to be absorbed into the learned transcriptomic programs. We therefore evaluated whether the DRGP method could recover and distinguish the genetic and transcriptional contributions to disease risk in the bulk GTEx simulations.

The recovered genetic effect increased monotonically with the simulated genetic liability across all transcriptomic settings, closely tracking the expected trend (Fig. 5A). Meanwhile,

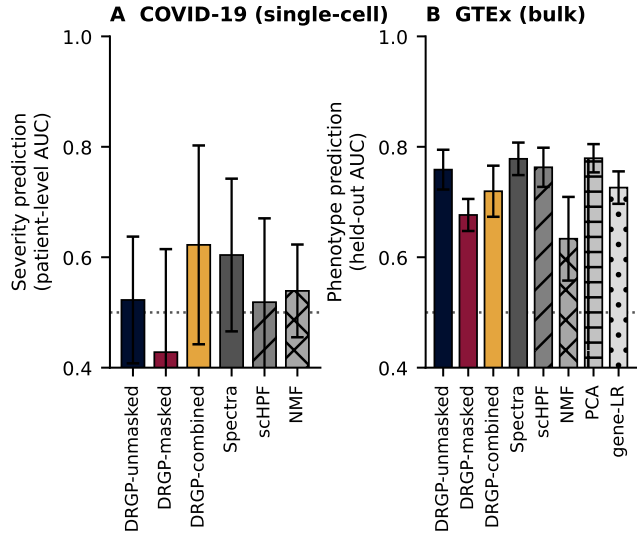


Fig. 4. **The DRGP method remains competitive with baselines on simulated RNA-seq DRP.** Left: COVID-19 scRNA-seq simulations evaluated on cells from held-out patients using inherited labels, aggregated to form patient-level prediction. Right: GTEx whole-blood simulations evaluated on held-out subjects. Bars show means with error bars denoting ± 1 SD. Dotted lines mark chance performance.

correlations between inferred gene programs and the PRS remained near the null (200 PRS permutations), consistent with their independence in the simulations (Fig. 5B). Held-out AUC also increased with w_s by 0.08–0.10 across all transcriptomic liability settings (Fig. 5C), demonstrating that genetic features complement expression-based DRP. Together, these results indicate that DRGP attributes the majority of genetic signal to the auxiliary component while maintaining limited residual association between the DRGPs and the PRS.

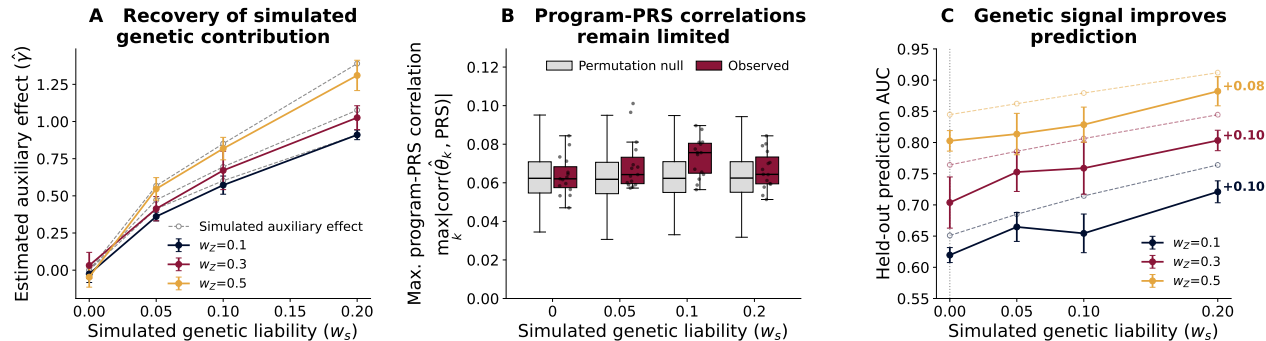


Fig. 5. The DRGP method separates simulated genetic liability from transcriptomic program activity. (A) Posterior mean PRS coefficient $\hat{y}$ versus the simulated genetic liability share w_s , stratified by transcriptomic liability share w_Z . The dashed line gives the simulated auxiliary effect; points and error bars give the mean ± 1 SD across 5 replicates. (B) Maximum absolute correlation between the PRS and any fitted program, $\max_k |\text{corr}(\hat{\theta}_k, \text{PRS})|$, compared with a permutation null. Boxes show the median and interquartile range, whiskers extend to $1.5 \times \text{IQR}$, and points denote replicates. (C) Held-out AUC as a function of w_s for each w_Z . Filled points and error bars show the mean ± 1 SD across five simulation replicates; light dashed curves show the maximum achievable AUC under the simulation. Labels report the increase in mean AUC from no genetic effect ($w_s = 0$) to a large effect ($w_s = 0.20$).

3.4. Experimental Results

COVID-19 severity prediction. We evaluated the patient-level test AUC as a function of the number of individuals (randomly chosen) for the scRNA-seq data (cells inherit the label of the individual) and a pseudo-bulk representation that sums the raw UMI counts across each individual’s cells (Fig. 6). We observed two unexpected trends in the scRNA-seq experimental data (Fig. 6A). Pathway-informed methods generally achieved lower AUC than pathway-agnostic methods (gene-LR, PCA, and DRGP-unmasked). AUC also decreased with the number of patients, possibly because the larger datasets contain more noisily labeled cells from which models learn spurious patterns. In contrast, pathway-informed methods performed better on pseudo-bulk data, and performance increased with sample size (Fig. 6B).

GTEx heart disease prediction. In the GTEx cohort, pathway information also appeared to reduce prediction performance (Fig. 6C). DRGP-unmasked achieved the highest mean predictive performance among gene program-based methods and was competitive with the best non-gene program methods; all factorization-based methods outperformed gene-LR, which is

traditionally used for DRP. Detailed hyperparameter settings for all DRGP experiments are listed in Table D2.

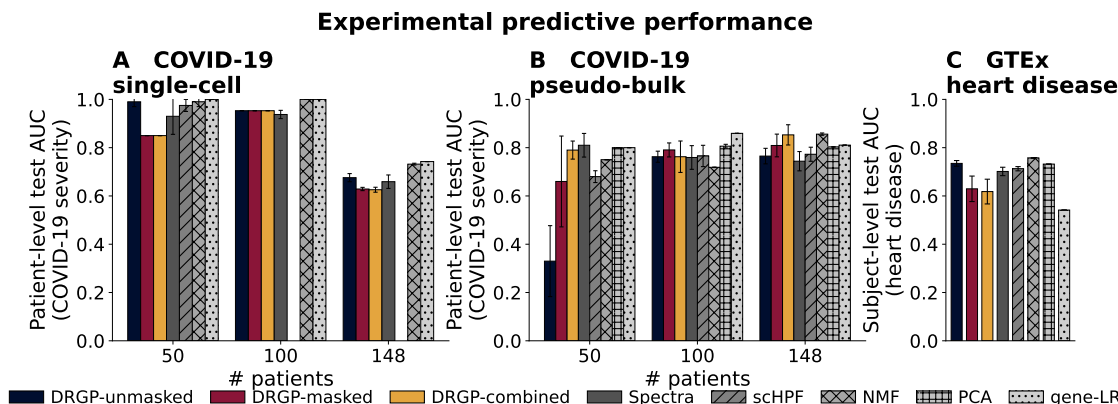


Fig. 6. **Predictive performance in experimental data.** Held-out AUC for (A) COVID-19 models fit to single cells, with cell-level probabilities averaged to produce a DRP for each patient; (B) COVID-19 models fit to patient-level pseudo-bulk profiles; and (C) GTEx whole-blood models for heart disease prediction. In (A) and (B), the x -axis gives the evaluated cohort size. Bars show means across a grouped train–test split and 5 random initializations; error bars show ± 1 SD. Missing bars indicate methods that exceeded available memory (500GB).

Interpretation of Learned Gene Programs. Since gene programs may vary across runs, we interpret only those stable across random initializations. For each cohort, we fit DRGP-combined with configurations listed in Table E1 on five random seeds and select programs that occurred in three or more of the five seeds with a mean pairwise Jaccard similarity of at least 0.30 between the top-50 genes (§E.3). Within programs, genes are ranked by $\hat{s}_{gl} = \mathbb{E}_q[\beta_{gl}]\mathbb{E}_q[\eta_g]$ (Eq. C.1), and programs are ordered by an effect-scaled importance score $I_\ell = |\bar{v}_{k\ell}|sd(\bar{\theta}_\ell)$ averaged over all seeds. Masked programs are more stable due to their pathway priors, but only 16/300 GTEx and 7/50 COVID-19 *de novo* programs were stable (Fig. E6). Appendix Table E2 summarizes supporting gene-function associations.

In the COVID-19 experiments, DRGP-combined identified gene programs representing critical cellular processes, including chemotaxis, protein translation, DNA replication, and microautophagy (Fig. E7). The program most strongly associated with severe disease (GP7) was composed of immune-related genes, including those involved in cell proliferation (11/50 genes), monocyte and macrophage markers (8/50 genes), and important immune-regulatory metallothionein proteins (7/50). Interestingly, the two top programs (GP7 and GP3) suggest metal ion homeostasis markers are important for predicting disease state, with seven metallothionein proteins as well as proteins that require binding to zinc or other metals to function (16/100). This is particularly relevant given the important role of metal ion homeostasis in infection susceptibility via immune cell function^{42,43}. Lastly, both GP7 and GP3 contain cytoskeletal genes, which may overlap with the top pathway, “Mechanical Load Activates Signaling By PIEZO1 and Integrins in Osteocytes”, which may suggest a role for cell recruitment and chemotaxis in prediction, despite opposite associations.

In the GTEx results, the pathways with the largest effect-scaled importance are critical cell processes including DNA repair, mRNA transcription, and mRNA decay (Fig. E8). The leading programs include several novel transcripts and pseudogenes, which may be related to differences in the central cell process pathways that are top predictors. In addition, GP13 has 24/50 membrane receptors, including 5 involved in cell adhesion, which has a known role in atherogenesis and plaque progression⁴⁴. GP3 and GP5 are enriched in markers for megakaryocytes/platelets and B cells, respectively. Platelets are crucial to vascular inflammation and vessel integrity^{45,46} while B cell antibodies can be either progressive or protective in atherogenesis depending on the epitopes targeted⁴⁷. Overall, our model combines program-level interpretability with substantially faster training than the baseline methods (Fig. F1).

4. Discussion

Several limitations and directions for future work warrant consideration. Learned programs represent disease-associated expression patterns rather than causal mechanisms, and pathway-informed interpretation depends on the quality and relevance of available annotations. In scRNA-seq, patient-level labels are inherited by individual cells even though disease-associated activity may be restricted to particular cell populations; patient-grouped splits and patient-level evaluation do not explicitly model this hierarchy. Future work should develop probabilistic pathway priors and hierarchical models that distinguish patient-, cell-type-, and cell-level variation. Additionally, while our DRGP model naturally extends to non-binary outcomes, the current evaluation is limited to binary outcomes and two transcriptomic cohorts.

The results suggested that strict pathway constraints can be limiting when curated annotations are incomplete or poorly matched to the disease or cellular context, motivating pathway priors that are adaptive or specific to cell-type or disease contexts. Longitudinal, prospective, and external-cohort studies could test whether learned programs generalize and precede disease progression, as our observational data cannot determine whether transcriptional changes cause disease or result from it. Additionally, fairness, calibration, and robustness under data distribution shift should be evaluated before clinical use. Because transcriptomic and auxiliary data may contain identifying information, secure data governance and access controls remain necessary to mitigate re-identification and unauthorized use risks^{48–50}.

5. Conclusions

In conclusion, we developed the DRGP model, which combines sparse supervised Bayesian Poisson factorization with regularized DRP, allowing gene programs, genetic and other covariates, and pathway knowledge to be modeled within a single interpretable framework. Across simulated datasets and experimental COVID-19 PBMC and GTEx whole-blood data, the DRGP method achieved the best predictive performance in some settings and remained competitive in others while providing program-level interpretations of disease risk. Simulations further showed that DRGP can recover disease-relevant transcriptional structure and use complementary genetic information without substantially encoding it in the learned gene programs. These findings support supervised gene-program learning as a flexible framework for interpretable disease-risk prediction from transcriptomic and auxiliary data.

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Appendix A. Additional Prior Work

Expression-based DRP requires feature selection to improve interpretability and reduce overfitting, in addition to supervised prediction via classification or survival analysis. *Wrapper* and *filter* algorithms use a predictive model or model-agnostic measure to score features, respectively; for example, AtheroSpectrum used wrapper-style search over random gene subsets to construct the CR-30 signature and then integrated this signature with clinical covariates¹. Wrapper methods are often better aligned with predictive objectives but are computationally expensive and can be unstable in high-dimensional regimes, whereas filter methods are computationally efficient but can be weakly coupled to the final prediction task. Embedded methods, such as L1-penalized logistic regression, optimize prediction and feature sparsity jointly and are therefore widely used in transcriptomic prediction pipelines. Because of their interpretability, generalized linear models are widely adopted to estimate disease risk^{2–4}; for instance, the coefficients of a logistic regression provide an interpretation of how gene expression changes affect the disease risk log-odds. Machine learning models like random forests or boosting classifiers have also been used for DRP^{5,6} and allow for learning interactions between features⁷, though these approaches are generally considered less interpretable than linear models and thus potentially less appropriate for clinical use⁸. Further, since machine learning models typically require more samples than linear regressors and most prior expression-based DRP studies have small sample sizes, gene expression is often imputed from genetic or epigenetic data⁹; however, gene expression is difficult to predict from genomics data¹⁰, with a small proportion of genes being able to be predicted with $r^2 > 0.1$ ^{11,12}. Across these approaches, the simplifying assumptions of additive linear effects^{4,13} or unstructured interactions among gene expression features^{5,6} poorly align with the coordinated gene activity driving disease-relevant cellular processes.

Appendix B. Variational Inference

The Poisson factorization, spike-and-slab structure, and supervised logistic term produce a coupled posterior over all latent variables, so exact inference is intractable. We use mean-field variational inference with coordinate-ascent updates, rendering the supervised term conditionally quadratic by Pólya–Gamma (PG) data augmentation¹⁴.

Appendix B.1. Pólya–Gamma Augmentation of the Supervised Term.

The only non-conjugate component of the model is the Bernoulli supervision

$$p(y_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) = \sigma(A_{jk})^{y_{jk}} (1 - \sigma(A_{jk}))^{1-y_{jk}}, \quad A_{jk} = \theta_j v_k^\top + \mathbf{x}_j^{\text{aux}} \gamma_k^\top,$$

where σ is the logistic function and $\theta_j, v_k, \mathbf{x}_j^{\text{aux}}, \gamma_k$ are row vectors. Following Polson, Scott, and Windle¹⁴, for a Pólya–Gamma variable $\zeta \sim \text{PG}(1, 0)$,

$$\frac{(e^A)^y}{1 + e^A} = \frac{1}{2} e^{\tilde{\kappa} A} \int_0^\infty e^{-\zeta A^2/2} p_{\text{PG}}(\zeta \mid 1, 0) d\zeta, \quad \tilde{\kappa} = y - \frac{1}{2}. \quad (\text{B.1})$$

Introducing one auxiliary variable ζ_{jk} per sample–phenotype pair gives the augmented likeli-

hood

$$p(y_{jk}, \zeta_{jk} \mid A_{jk}) \propto \exp\left(\tilde{\kappa}_{jk} A_{jk} - \frac{\zeta_{jk}}{2} A_{jk}^2\right) p_{\text{PG}}(\zeta_{jk} \mid 1, 0), \quad \tilde{\kappa}_{jk} = y_{jk} - \frac{1}{2}. \quad (\text{B.2})$$

We write the PG shift as $\tilde{\kappa}_{jk}$ to distinguish it from the number of phenotypes κ . Equation (B.1) is an exact identity rather than a variational bound, so the augmented model has the same marginal likelihood as the original. Conditional on ζ_{jk} the exponent is quadratic in A_{jk} , which is conjugate to the Gaussian factors $q(v_k)$ and $q(\gamma_k)$ but only log-quadratic in θ_j ; the consequences for the θ update are given in the remark below.

Appendix B.2. Joint Distribution and Variational Family.

Conditioned on the auxiliary covariates and including the augmentation variables, the joint distribution factorizes as

$$\begin{aligned} p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, M, \pi, \eta, \mathbf{v}, s, \boldsymbol{\gamma}, \boldsymbol{\zeta} \mid \mathbf{X}^{\text{aux}}) &= \prod_{g=1}^p \left[p(\pi_g) p(\eta_g) \prod_{\ell=1}^d p(m_{g\ell} \mid \pi_g) p(\tilde{\beta}_{g\ell} \mid \eta_g) \right] \\ &\times \prod_{j=1}^n \left[p(\xi_j) \prod_{\ell=1}^d p(\theta_{j\ell} \mid \xi_j) \prod_{g=1}^p p(x_{jg} \mid \theta_j, \beta_g) \right] \\ &\times \prod_{k=1}^{\kappa} \left[p(\gamma_k) \prod_{\ell=1}^d p(s_{k\ell}) p(v_{k\ell} \mid s_{k\ell}) \prod_{j=1}^n p(y_{jk}, \zeta_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) \right]. \end{aligned}$$

We assume the fully factorized variational family

$$\begin{aligned} q(\eta_g) &= \text{Gamma}(a_{\eta_g}, b_{\eta_g}), & q(\tilde{\beta}_{g\ell}) &= \text{Gamma}(a_{\tilde{\beta}_{g\ell}}, b_{\tilde{\beta}_{g\ell}}), \\ q(m_{g\ell}) &= \text{Bernoulli}(\rho_{g\ell}), & q(\pi_g) &= \text{Beta}(a_{\pi_g}, b_{\pi_g}), \\ q(\xi_j) &= \text{Gamma}(a_{\xi_j}, b_{\xi_j}), & q(\theta_{j\ell}) &= \text{Gamma}(a_{\theta_{j\ell}}, b_{\theta_{j\ell}}), \\ q(v_{k\ell}) &= \mathcal{N}(\mu_{v_{k\ell}}, \tau_{v_{k\ell}}^2), & q(s_{k\ell}^{-1}) &= \text{InvGaussian}(\mu_{s_{k\ell}}, \lambda_{s_{k\ell}}), \\ q(\gamma_k) &= \mathcal{N}(\mu_{\gamma_k}, \Sigma_{\gamma_k}), & q(\zeta_{jk}) &= \text{PG}(1, c_{jk}), \end{aligned}$$

where the Bayesian-Lasso factor is placed on the *precision* $s_{k\ell}^{-1}$, which is inverse-Gaussian under the Gaussian-exponential scale mixture. For any Gamma factor $q(X) = \text{Gamma}(a, b)$ we use $\mathbb{E}_q[X] = a/b$, $\mathbb{E}_q[\log X] = \Psi(a) - \log b$, and $\text{Var}_q(X) = a/b^2$, where $\Psi(\cdot)$ is the digamma function.

Optimal augmentation factor. Because the augmented exponent in (B.2) is $-\frac{1}{2}\zeta_{jk}A_{jk}^2$, the coordinate-optimal factor for ζ_{jk} is again Pólya-Gamma, exponentially tilted with $c_{jk} = \sqrt{\mathbb{E}_q[A_{jk}^2]}$. The only moment used downstream is its mean,

$$\bar{\zeta}_{jk} \equiv \mathbb{E}_q[\zeta_{jk}] = \frac{1}{2c_{jk}} \tanh\left(\frac{c_{jk}}{2}\right), \quad \lim_{c_{jk} \rightarrow 0} \bar{\zeta}_{jk} = \frac{1}{4}, \quad (\text{B.3})$$

evaluated by a second-order Taylor expansion of $\tanh$ when c_{jk} is small.

Count allocation and gene-side updates. Following Raikov's decomposition theorem¹⁵, we introduce latent allocations $z_{jg\ell}$ with $x_{jg} = \sum_{\ell=1}^d z_{jg\ell}$ and $z_{jg\ell} \sim \text{Poisson}(\theta_{j\ell}\beta_{g\ell})$. Because

$\beta_{g\ell} = m_{g\ell}\tilde{\beta}_{g\ell}$, inactive gene-program pairs contribute zero rate automatically. Writing $e_{jg\ell} = \exp(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}})$, the optimal multinomial allocation is

$$\phi_{jg\ell} = \frac{\rho_{g\ell} e_{jg\ell}}{\sum_{\ell'=1}^d \rho_{g\ell'} e_{jg\ell'}}, \quad \mathbb{E}_q[z_{jg\ell}] = x_{jg}\phi_{jg\ell}, \quad (\text{B.4})$$

and the slab loading update is

$$a_{\tilde{\beta}_{g\ell}} = \alpha_\beta + \sum_{j=1}^n x_{jg}\phi_{jg\ell}, \quad b_{\tilde{\beta}_{g\ell}} = \mathbb{E}_q[\eta_g] + \rho_{g\ell} \sum_{j=1}^n \mathbb{E}_q[\theta_{j\ell}]. \quad (\text{B.5})$$

For the Bernoulli membership variable, the log-odds update compares the expected log-joint under $m_{g\ell} = 1$ and $m_{g\ell} = 0$. Let $\bar{\phi}_{jg\ell} = e_{jg\ell}(e_{jg\ell} + \sum_{m \neq \ell} \rho_{gm} e_{jgm})^{-1}$ be the allocation with program ℓ held active, and define

$$L_{jg\ell} = x_{jg} \bar{\phi}_{jg\ell} \left(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}} \right) - \mathbb{E}_q[\theta_{j\ell}] \mathbb{E}_q[\tilde{\beta}_{g\ell}] - \log \Gamma(x_{jg} \bar{\phi}_{jg\ell} + 1);$$

the final term evaluates $\mathbb{E}_q[\log z_{jg\ell}!]$ at $\mathbb{E}_q[z_{jg\ell}]$ rather than exactly. The update is

$$\log \frac{\rho_{g\ell}}{1 - \rho_{g\ell}} = \Psi(a_{\pi_g}) - \Psi(b_{\pi_g}) + \sum_{j=1}^n L_{jg\ell}. \quad (\text{B.6})$$

The conjugate hyperparameter updates are

$$a_{\pi_g} = \alpha_\pi + \sum_{\ell=1}^d \rho_{g\ell}, \quad b_{\pi_g} = \beta_\pi + d - \sum_{\ell=1}^d \rho_{g\ell}, \quad (\text{B.7})$$

$$a_{\eta_g} = \alpha_\eta + d\alpha_\beta, \quad b_{\eta_g} = \lambda_\eta + \sum_{\ell=1}^d \frac{a_{\tilde{\beta}_{g\ell}}}{b_{\tilde{\beta}_{g\ell}}}, \quad a_{\xi_j} = \alpha_\xi + d\alpha_\theta, \quad b_{\xi_j} = \lambda_\xi + \sum_{\ell=1}^d \frac{a_{\theta_{j\ell}}}{b_{\theta_{j\ell}}}. \quad (\text{B.8})$$

Pathway mask. In the de novo setting all gene-program pairs are updated freely. In the pathway-guided setting we fix $m_{g\ell} = 0$, $\rho_{g\ell} = 0$, and $\beta_{g\ell} = 0$ for every entry with $M_{g\ell} = 0$, which implies $\phi_{jg\ell} = 0$ for all j ; the same updates are applied to the allowed entries only. The gene-side sums over ℓ in (B.7)–(B.8) run over allowed entries, so d is replaced by $d_g = \sum_\ell M_{g\ell}$ in those three updates. The sample-side count d in a_{ξ_j} is unchanged, since θ is not masked.

Supervised updates under augmentation. Conditional on $q(\zeta)$, the augmented supervised term (B.2) is quadratic in A_{jk} , and $\bar{\zeta}_{jk}$ plays the role of the quadratic-curvature coefficient. Throughout we use the leave-one-out predictor $C_{jk}^{(-\ell)} = \sum_{m \neq \ell} \mathbb{E}_q[\theta_{jm}] \mathbb{E}_q[v_{km}] + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}$, i.e. A_{jk} with the ℓ -th program term removed. The sample-activity update is

$$a_{\theta_{j\ell}} = \alpha_\theta + \sum_{g=1}^p x_{jg}\phi_{jg\ell}, \quad b_{\theta_{j\ell}} = \mathbb{E}_q[\xi_j] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}] + R_{j\ell}, \quad (\text{B.9})$$

$$R_{j\ell} = \sum_{k=1}^{\kappa} \left[-\bar{\kappa}_{jk} \mathbb{E}_q[v_{k\ell}] + \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}] C_{jk}^{(-\ell)} + \frac{1}{2} \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2] \mathbb{E}_q[\theta_{j\ell}] \right]. \quad (\text{B.10})$$

The supervised correction $R_{j\ell}$ enters the Gamma rate of $\theta_{j\ell}$ and couples program activity to disease status: the label-linear term shifts disease-relevant programs ($v_{k\ell} \neq 0$) toward higher

activity in cases and lower activity in controls, while the $\bar{\zeta}_{jk}$ -scaled terms damp that shift by the curvature of the logistic.

Remark (exactness of the θ update). Conditional on ζ , the terms in $\theta_{j\ell}$ contributed by (B.2) are $\tilde{\kappa}_{jk} v_{k\ell} \theta_{j\ell} - \bar{\zeta}_{jk} v_{k\ell} C_{jk}^{(-\ell)} \theta_{j\ell} - \frac{1}{2} \bar{\zeta}_{jk} v_{k\ell}^2 \theta_{j\ell}^2$, so the coordinate-optimal factor is $q^*(\theta_{j\ell}) \propto \theta_{j\ell}^{a-1} \exp(-b\theta_{j\ell} - c\theta_{j\ell}^2)$ with $c = \frac{1}{2} \sum_k \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2]$. The quadratic term lies outside the span of the Gamma sufficient statistics $(\log \theta, \theta)$, so q^* is not Gamma. Equation (B.9) therefore restricts $q(\theta_{j\ell})$ to the Gamma family and linearizes $c\theta_{j\ell}^2$ about the current mean, replacing it with $(c\mathbb{E}_q[\theta_{j\ell}])\theta_{j\ell}$; the resulting update is a fixed-point iteration rather than an exact coordinate maximizer. Because the linearized rate may be negative, it is floored at a small $\varepsilon > 0$, $b_{\theta_{j\ell}} \leftarrow \max(b_{\theta_{j\ell}}, \varepsilon)$, which is a projection back into the family.

Regression and scale updates. The regression coefficient updates are

$$\tau_{v_{k\ell}}^{-2} = \mathbb{E}_q[s_{k\ell}^{-1}] + \sum_{j=1}^n \bar{\zeta}_{jk} \mathbb{E}_q[\theta_{j\ell}^2], \quad \mu_{v_{k\ell}} = \tau_{v_{k\ell}}^2 \sum_{j=1}^n \left[\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} C_{jk}^{(-\ell)} \right] \mathbb{E}_q[\theta_{j\ell}], \quad (\text{B.11})$$

with $\mathbb{E}_q[\theta_{j\ell}^2] = \mathbb{E}_q[\theta_{j\ell}]^2 + \text{Var}_q(\theta_{j\ell})$, and

$$\Sigma_{\gamma_k}^{-1} = \frac{1}{\sigma_{\gamma}^2} I + \sum_{j=1}^n \bar{\zeta}_{jk} (\mathbf{x}_j^{\text{aux}})^{\top} \mathbf{x}_j^{\text{aux}}, \quad \mu_{\gamma_k} = \Sigma_{\gamma_k} \sum_{j=1}^n (\mathbf{x}_j^{\text{aux}})^{\top} \left[\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} \mathbb{E}_q[\theta_j] \mathbb{E}_q[v_k] \right]^{\top}. \quad (\text{B.12})$$

The Bayesian-Lasso scale update is unchanged by the augmentation, since the prior on v is untouched. Under $v_{k\ell} \mid s_{k\ell} \sim \mathcal{N}(0, s_{k\ell})$, $s_{k\ell} \sim \text{Exp}(1/2b_v^2)$, the optimal factor for $s_{k\ell}^{-1}$ is inverse-Gaussian with $\lambda_{s_{k\ell}} = b_v^{-2}$ and

$$\mathbb{E}_q[s_{k\ell}^{-1}] = \mu_{s_{k\ell}} = \frac{1}{b_v \sqrt{\mathbb{E}_q[v_{k\ell}^2]}}. \quad (\text{B.13})$$

Finally, the tilt is refreshed by setting $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because the variational factors are independent, $\text{Var}_q(A_{jk})$ decomposes into separate θ -, v -, and γ -contributions,

$$\begin{aligned} \mathbb{E}_q[A_{jk}] &= \mathbb{E}_q[\theta_j] \mu_{v_k}^{\top} + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}, \\ c_{jk}^2 &= \mathbb{E}_q[A_{jk}^2] = \mathbb{E}_q[A_{jk}]^2 + \sum_{\ell=1}^d \left[\text{Var}_q(\theta_{j\ell}) \mathbb{E}_q[v_{k\ell}^2] + \mathbb{E}_q[\theta_{j\ell}]^2 \tau_{v_{k\ell}}^2 \right] + \mathbf{x}_j^{\text{aux}} \Sigma_{\gamma_k} (\mathbf{x}_j^{\text{aux}})^{\top}, \end{aligned} \quad (\text{B.14})$$

with $\mathbb{E}_q[v_{k\ell}^2] = \mu_{v_{k\ell}}^2 + \tau_{v_{k\ell}}^2$, after which $\bar{\zeta}_{jk}$ is recomputed via (B.3). All three variance terms are required for the tilt to equal the full second moment, so that the residual in (B.15) vanishes at the optimum.

Objective and optimization. The variational objective is the evidence lower bound of the augmented model, $\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, \mathbf{M}, \boldsymbol{\pi}, \boldsymbol{\eta}, \mathbf{v}, \mathbf{s}, \boldsymbol{\gamma}, \boldsymbol{\zeta} \mid \mathbf{X}^{\text{aux}})] - \mathbb{E}_q[\log q]$, combining the expected log-priors, the augmented Poisson likelihood, the augmented supervised term, and the entropies of all factors. Marginalizing the optimal PG factor analytically, the supervised contribution at $q(\zeta_{jk}) = \text{PG}(1, c_{jk})$ reduces to

$$\mathcal{L}^{\text{sup}} = \sum_{j=1}^n \sum_{k=1}^{\kappa} \left[\tilde{\kappa}_{jk} \mathbb{E}_q[A_{jk}] - \frac{c_{jk}}{2} + \log \sigma(c_{jk}) - \frac{\bar{\zeta}_{jk}}{2} (\mathbb{E}_q[A_{jk}^2] - c_{jk}^2) \right], \quad (\text{B.15})$$

whose last term vanishes when the tilt is at its optimum $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because (B.1) is exact rather than a bound, $\mathcal{L}(q)$ is a valid lower bound on $\log p(\mathbf{X}, \mathbf{Y} \mid \mathbf{X}^{\text{aux}})$ at every iteration. Monotonicity holds for every block whose update is an exact coordinate maximizer, but not for θ , whose update is linearized; we therefore monitor the ELBO empirically and base model selection and early stopping on validation performance, with all reported predictive results evaluated on held-out data. Algorithm 2 summarizes the procedure.

Complexity. Let $\text{nnz}(\mathbf{X}) = |\{(j, g) : x_{jg} \neq 0\}|$. The allocation (B.4) is evaluated on the fly for each nonzero and never materialized, giving $O(\text{nnz}(\mathbf{X})d)$ time and $O(d)$ working memory per entry; the loading update (B.5) reuses the same sums. In (B.6) the term $\mathbb{E}_q[\theta_{j\ell}]\mathbb{E}_q[\tilde{\beta}_{g\ell}]$ is nonzero when $x_{jg} = 0$, but factorizes as $(\sum_j \mathbb{E}_q[\theta_{j\ell}])\mathbb{E}_q[\tilde{\beta}_{g\ell}]$, so its cost is $O(pd)$ rather than $O(npd)$. Accumulating $\sum_m \mathbb{E}_q[\theta_{jm}]\mathbb{E}_q[v_{km}]$ once per (j, k) and subtracting the ℓ -th term makes all $C_{jk}^{(-\ell)}$ available in $O(nd\kappa)$ rather than $O(nd^2\kappa)$, which also bounds (B.11) and (B.14); the auxiliary update (B.12) costs $O(n(p^{\text{aux}})^2 + (p^{\text{aux}})^3)$ per phenotype. A training sweep is therefore $O(\text{nnz}(\mathbf{X})d + nd\kappa + \kappa\{n(p^{\text{aux}})^2 + (p^{\text{aux}})^3\})$, dominated by the first term whenever $p^{\text{aux}} \ll p$. Memory is $O(\text{nnz}(\mathbf{X}) + (n + p)d + \kappa(d + (p^{\text{aux}})^2))$. Fold-in for one held-out sample costs $O(\text{nnz}(\mathbf{x}_{j^*})d)$ per local iteration and touches no global parameter, so test-time inference is embarrassingly parallel across samples.

Test- and validation-time inference (fold-in). At test time the labels y_{j^*k} are unobserved, so (B.2) contributes no likelihood factor for a new sample j^* . We freeze the *global* parameters—the gene-side factors $\{a_{\tilde{\beta}}, b_{\tilde{\beta}}, \rho, a_{\eta}, b_{\eta}, a_{\pi}, b_{\pi}\}$ and the regression coefficients $\{\mu_v, \tau_v^2, \mu_{\gamma}, \Sigma_{\gamma}\}$ —and infer only the *local* variables $\{\theta_{j^*\ell}, \xi_{j^*}\}$ from the counts x_{j^*g} through the allocations $\phi_{j^*g\ell}$. The supervised correction vanishes, $R_{j^*\ell} = 0$, no PG variables ζ_{j^*k} are needed, and (B.9) reduces to its unsupervised form

$$a_{\theta_{j^*\ell}} = \alpha_{\theta} + \sum_{g=1}^p x_{j^*g} \phi_{j^*g\ell}, \quad b_{\theta_{j^*\ell}} = \mathbb{E}_q[\xi_{j^*}] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}]. \quad (\text{B.16})$$

Note that θ_{j^*} is thus driven only by held-out expression and the fixed gene programs, whereas training activities are label-informed through $R_{j\ell}$. Given the converged fold-in posterior, the predictor moments are read off from (B.14) as $\mathbb{E}_q[A_{j^*k}] = \mathbb{E}_q[\theta_{j^*}]\mu_{v_k}^{\top} + \mathbf{x}_{j^*}^{\text{aux}}\mu_{\gamma_k}$ and $\text{Var}_q(A_{j^*k}) = c_{j^*k}^2 - \mathbb{E}_q[A_{j^*k}]^2$, and the posterior predictive probability is approximated by the Gaussian–logistic integral

$$\hat{y}_{j^*k} = \mathbb{E}_q[\sigma(A_{j^*k})] \approx \sigma\left(\frac{\mathbb{E}_q[A_{j^*k}]}{\sqrt{1 + \pi \text{Var}_q(A_{j^*k})/8}}\right), \quad (\text{B.17})$$

where π denotes the mathematical constant. This reduces to the plug-in estimate $\sigma(\mathbb{E}_q[A_{j^*k}])$ in the zero-variance limit. The same routine supplies the validation predictions used for model selection in Algorithm 2, and is summarized in Algorithm 3.

Algorithm 2 Pólya–Gamma CAVI for DRGP

Input: Expression $\mathbf{X}$, labels $\mathbf{Y}$, auxiliary covariates $\mathbf{X}^{\text{aux}}$, number of factors d , pathway mask $\mathbf{M}$, and model hyperparameters

Output: Fitted global and local variational parameters

- 1: Initialize all variational parameters and enforce $\mathbf{M}$
 - 2: **while** the ELBO has not converged **do**
 - 3: Update count allocations ϕ ▷ Eq. (B.4)
 - 4: Update gene loadings and memberships $(\tilde{\beta}, \rho)$ ▷ Eqs. (B.5), (B.6)
 - 5: Update gene-level parameters (π, η) ▷ Eq. (B.7), (B.8)
 - 6: Update sample activities (θ, ξ) ▷ Eqs. (B.9), (B.8)
 - 7: Update supervised parameters (s, v, γ) ▷ Eqs. (B.13)–(B.12)
 - 8: Update Pólya–Gamma moments c and $\bar{\zeta}$ ▷ Eqs. (B.14), (B.3)
 - 9: **end while**
 - 10: **return** fitted variational parameters
-

Algorithm 3 Fold-in inference and risk prediction

Input: Held-out expression $\mathbf{x}_{j^*}$, auxiliary covariates $\mathbf{x}_{j^*}^{\text{aux}}$, and frozen global parameters from Algorithm 2

Output: Predicted risks $\hat{y}_{j^*k}$

- 1: **for** each held-out sample j^* **do**
 - 2: Initialize local parameters $(\phi_{j^*}, \theta_{j^*}, \xi_{j^*})$
 - 3: **while** the local Poisson objective has not converged **do**
 - 4: Update ϕ_{j^*} , then θ_{j^*} with $R_{j^*\ell} = 0$, then ξ_{j^*} ▷ Eqs. (B.4), (B.16), (B.8)
 - 5: **end while**
 - 6: **for** each phenotype k **do**
 - 7: Compute posterior moments of A_{j^*k} and predict $\hat{y}_{j^*k}$ ▷ Eqs. (B.14), (B.17)
 - 8: **end for**
 - 9: **end for**
 - 10: **return** $\{\hat{y}_{j^*k}\}$
-

Appendix C. Ranking Genes within DRGP Programs

For all downstream analyses requiring a continuous ranking of genes within a DRGP program, we use

$$\hat{s}_{g\ell} = \mathbb{E}_q[\beta_{g\ell}] \mathbb{E}_q[\eta_g]. \quad (\text{C.1})$$

The expected loading $\mathbb{E}_q[\beta_{g\ell}]$ incorporates the posterior gene–program inclusion probability, while $\mathbb{E}_q[\eta_g]$ adjusts for gene-specific loading capacity. Consequently, $\hat{s}_{g\ell}$ favors genes that load strongly on program ℓ but do not load broadly across many programs.

For biological program summaries, genes were ordered by $\hat{s}_{g\ell}$ and the top 50 genes were retained where a fixed-size list was required. In simulation, support recovery was evaluated using $\hat{s}_{\cdot\ell}$ as the continuous gene score, with fitted programs Hungarian-matched to simulated programs by signed cosine similarity of the gene-score vectors. Program reproducibility across

random initializations was likewise evaluated by cosine similarity between gene-score vectors.

Appendix D. Hyperparameter selection

For simulations, hyperparameters were fixed before evaluation and held constant across all simulated difficulty settings.

Table D1. Fixed parameters of the simulation framework. Swept parameters (r , w_Z , ρ , δ , φ , w_s) are given in §3.2.

Symbol	Meaning	Single-cell	Bulk
–	individuals / samples	80 (100 cells each)	2,000
p	genes	10,000	5,000
–	cell types	6	–
–	disease programs (annot. + <i>de novo</i>)	2 + 2	2 + 4
–	nuisance programs (annot. + <i>de novo</i>)	0 + 1	1 + 1
$ S_\ell $	program support size	U{50,100}	U{50,150}
α_p, β_p	propensity Beta prior	2.0, 2.0	–
σ_u	per-program propensity jitter	0.5	–
(q_0, q_1)	baseline / max carrier prob.	(0.20, 0.80)	–
λ_ℓ	program prevalence multiplier	1.0 (all ℓ)	–
π_ℓ	nuisance carrier rate	0.5	0.5
ω_ℓ	liability weight per DRGP	{2.0, 0.7, 2.0, 0.7}, balanced signs	$ v = 1.5$
–	nuisance liability weight	0	0
–	liability form / prevalence	probit / 0.5	probit / 0.26
–	activation magnitude	Gamma(2,2)	binary carrier
–	program polarity	up/up/mixed/down/mixed	up
–	truth draws per configuration	5	5

For experimental analyses, hyperparameters were selected separately for each dataset, input representation, cohort size, and prior-knowledge mode using tree-structured Parzen estimator optimization implemented in Optuna¹⁶. The objective was mean validation ROC-AUC. For single-cell analyses, all cells from the same patient were assigned to the same partition to prevent patient-level information leakage between training and validation sets. We evaluated 30 trials for each COVID-19 single-cell configuration and 60 trials for each COVID-19 pseudo-bulk and GTEx configuration.

The search included the number of latent programs K ; the Gamma prior shapes α_θ , α_β , α_ξ , and α_η ; the Laplace scale b_v for the supervised program coefficients; the prior standard deviation σ_γ for auxiliary-covariate coefficients; and the spike-and-slab parameters α_π and $\beta_{\pi, \text{scale}}$. For pathway-masked models, K was fixed to the number of supplied pathways. For combined models, the pathway-constrained factors were fixed and the number of additional *de novo* factors was optimized. The configuration with the highest validation AUC was used for the final fit. Final experimental models were run for at most 500 coordinate-ascent iterations with an ELBO-based relative convergence tolerance of 10^{-3} and a 50-iteration warm-up before updating the supervised coefficients. The selected values for each reported configuration are provided in Table D2.

In contrast, simulation hyperparameters were fixed before evaluation and held constant across all simulated difficulty settings. This design prevents changes in model tuning from being confounded with changes in perturbation strength, responder-cell prevalence, or the transcriptomic and auxiliary liability shares. Both simulation arms used $\alpha_\theta = \alpha_\beta = 0.3$, $\sigma_\gamma = 1$, a convergence tolerance of 10^{-3} , and a maximum of 3,000 coordinate-ascent iterations. The COVID-19 single-cell simulations were fit at each of $d \in \{8, 10, 14\}$ latent programs, with the supervised-coefficient scale held fixed at $b_v = 1$. In the GTEx bulk simulations the number of simulated programs is $d^* = 8$, while the fitted models used $d = 10$ for the unmasked configuration, $d_{\text{path}} = 3$ for the pathway-masked configuration, and 3 pathway-anchored plus 7 de novo factors for the combined configuration; all unsupervised baselines were fit with $d = 10$ for comparability. In this arm b_v was initialized at 1 and auto-calibrated during fitting to the data precision, converging to ≈ 0.01 .

All comparison methods were run with default hyperparameters.

Table D2. **Selected hyperparameters for the experimental configurations.** Values maximize patient- or subject-grouped validation AUC. The number of latent gene programs is written $K_{\text{pathway}} + K_{\text{de novo}}$: unmasked models are entirely *de novo*, masked models are entirely pathway-anchored, and combined models use both.

Dataset/representation	Mode	$K_{\text{pathway}} + K_{\text{de novo}}$	α_θ	α_β	α_ξ	α_η	b_v	σ_γ	α_π	$\beta_{\pi, \text{scale}}$
COVID-19 single-cell	Unmasked	0 + 50	3.0266	2.3343	2.3163	1.7338	0.0639	0.1205	2.7179	1.5597
COVID-19 single-cell	Masked	346 + 0	2.0708	1.0016	2.1426	1.6326	0.2451	1.4332	0.1292	17.2617
COVID-19 single-cell	Combined	346 + 50	2.0708	1.0016	2.1426	1.6326	0.2451	1.4332	0.1292	17.2617
COVID-19 pseudo-bulk	Unmasked	0 + 25	2.3066	0.9487	3.9071	0.8187	0.0932	0.7832	1.8736	8.4993
COVID-19 pseudo-bulk	Masked	351 + 0	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
COVID-19 pseudo-bulk	Combined	351 + 50	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
GTEx whole blood	Unmasked	0 + 20	3.3541	2.9740	3.0515	3.1994	0.0657	0.2927	0.1573	29.2676
GTEx whole blood	Masked	309 + 0	0.7029	1.4259	0.6661	1.2495	0.8394	1.5575	0.5051	10.4172
GTEx whole blood	Combined	309 + 20	0.7029	1.4259	0.6661	1.2495	0.8394	1.5575	0.5051	10.4172

Appendix E. Additional Results

Appendix E.1. *Simulation Results Across the Parameter Sweep*

The single-cell simulation was run over the difficulty grid of §3.2 rather than a single configuration, summarised here as marginal profiles along each factor. Three patterns hold across the sweeps. Annotated recovery is set by whether a method is given the supports rather than by the difficulty of the instance: DRGP-masked is invariant at 0.801 and DRGP-combined spans 0.762–0.801, while methods without the supports stay near the chance level $|S_\ell|/p = 0.0075$ (Figs. E1, E2). De novo recovery is at chance for every method in every configuration (0.008–0.010). Patient-level prediction responds most to the responder fraction and to the transcriptomic liability share, and least to the perturbation strength: across the sweep the largest swing in AUC is 0.146 for ρ and 0.118 for w_Z , against at most 0.092 for r (Figs. E4, E3).

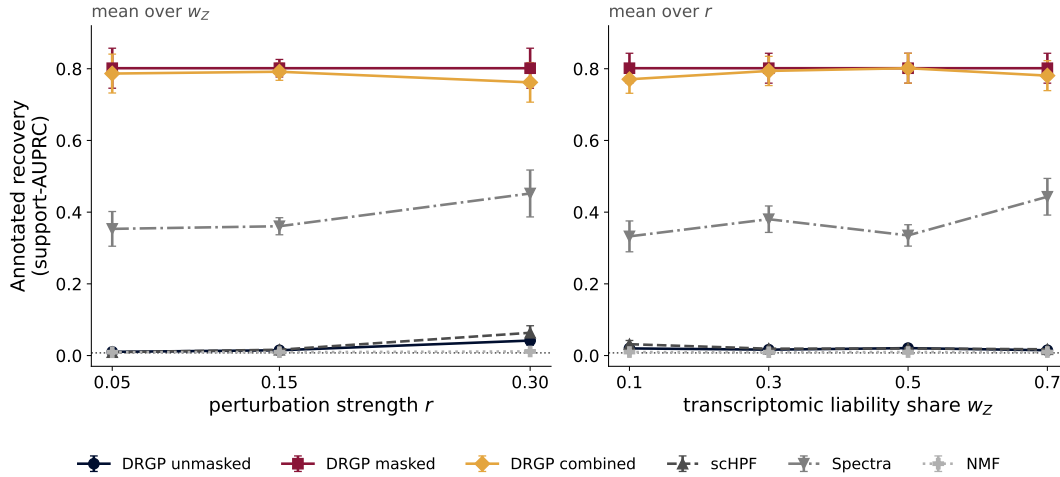


Fig. E1. **Recovery of Pathways (support-AUPRC) as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: Support-AUPRC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: Support-AUPRC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold $\rho = 0.3$ and $d = 8$. Points are means over the five simulated truth draws with ± 1 standard error, and the dotted line marks chance ($|S_\ell|/p = 0.0075$).

Appendix E.2. *Cell-Level Prediction in the Single-Cell Simulation*

Assigning each cell its individual’s phenotype produces a label that individual cells cannot support. At the headline setting of the simulated grid ($w_Z = 0.7$, $r = 0.30$), cell-level prediction remains at chance for every method (0.50–0.52 AUC, Fig. E5, not because the programs are undetectable, but because a responder fraction of $\rho = 0.3$ leaves most cells in a case individual carrying no DRGP at all. Averaging a patient’s cell-level probabilities recovers the signal that individual cells cannot resolve, making patient-level performance the more appropriate and reliable measure of predictive accuracy in this setting.

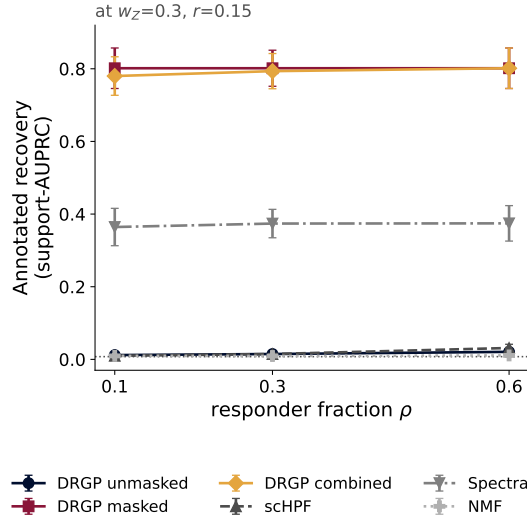


Fig. E2. **Recovery of Pathways (support-AUPRC) on the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, $r = 0.15$, $d = 8$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance, the mean support prevalence $|S_\ell|/p = 0.0075$.

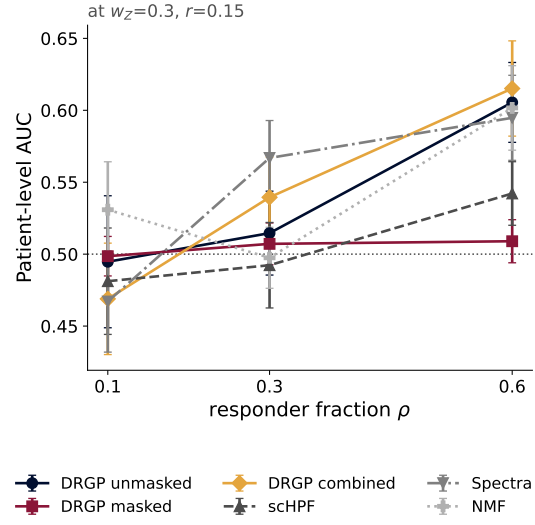


Fig. E3. **Patient-level AUC in the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, and $r = 0.15$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance.

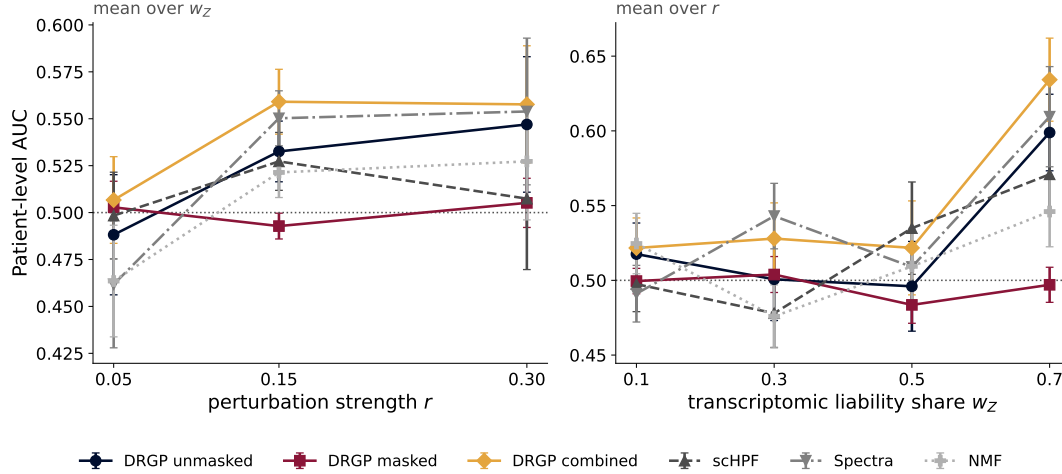


Fig. E4. **Patient-level AUC in the COVID-19 single-cell simulation as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: AUC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: AUC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold the responder fraction at $\rho = 0.3$ and the latent dimension at $d = 8$. Points are means over the five simulated truth draws per each configuration, and error bars show ± 1 standard error. The dotted line marks chance AUC = 0.5.

Appendix E.3. Reproducible Gene Programs on Experimental Results

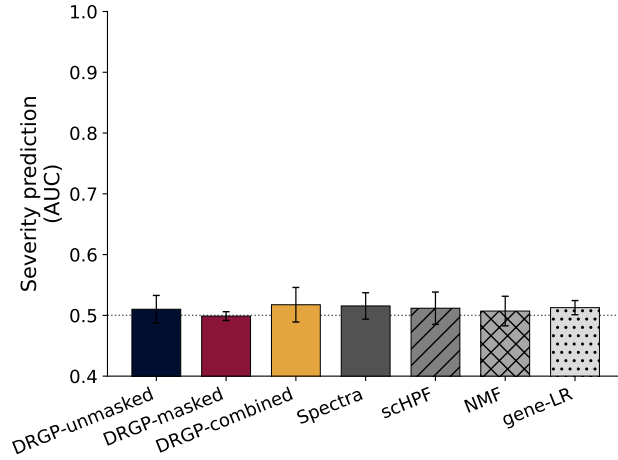


Fig. E5. **All methods remain at chance on simulated DRP.** COVID-19 scRNA-seq simulations evaluated on cells from held-out patients using inherited labels. Bars show means with error bars denoting ± 1 SD. Dotted lines mark chance performance.

Table E1. **Configurations used for program interpretation.** Both cohorts were fit in *combined* mode with five random seeds, varying only the model-initialization seed. d is written $d_{\text{path}} + d_{\text{de novo}}$. Values not listed were left at their defaults; in particular α_ξ and α_η were not tuned for these runs.

	GTEX whole blood	COVID-19 pseudo-bulk
Label	heart disease	severity
Input	bulk counts, 27,693 genes	pseudo-bulk counts, 148 patients
Pathway mask	Reactome curated (depth 2, ≥ 15)	Reactome immune
$K_{\text{pathway}} + K_{\text{de novo}}$	309 + 300	358 + 50
Auxiliary covariates	5 (sex, BMI, smoking, hypertension, T2D)	sex, 3 comorbidities
α_θ	0.6814	2.4135
α_β	2.0062	1.1470
α_ξ	1.0 (default)	1.0 (default)
α_η	1.0 (default)	1.0 (default)
b_v	0.2413	1.5999
σ_γ	1.2273	1.4595
α_π	0.2407	1.0371
$\beta_{\pi, \text{scale}}$	5.1838	36.8330
Max. CAVI iterations	500	500
Relative tolerance	10^{-3}	10^{-3}
Supervised warm-up	50 iterations	50 iterations
Early stopping	ELBO	ELBO
Regression design	normalized	standard

The programs interpreted in §3.4 were selected by a cross-seed reproducibility filter, applied separately to each cohort.

For interpretation, we use a deliberately larger program budget than prediction: the tuned predictive models allocate only 20 (GTEX) and 50 (COVID-19) free factors, which is enough to maximize held-out AUC but too few to separate distinct biological signals, so we refit each

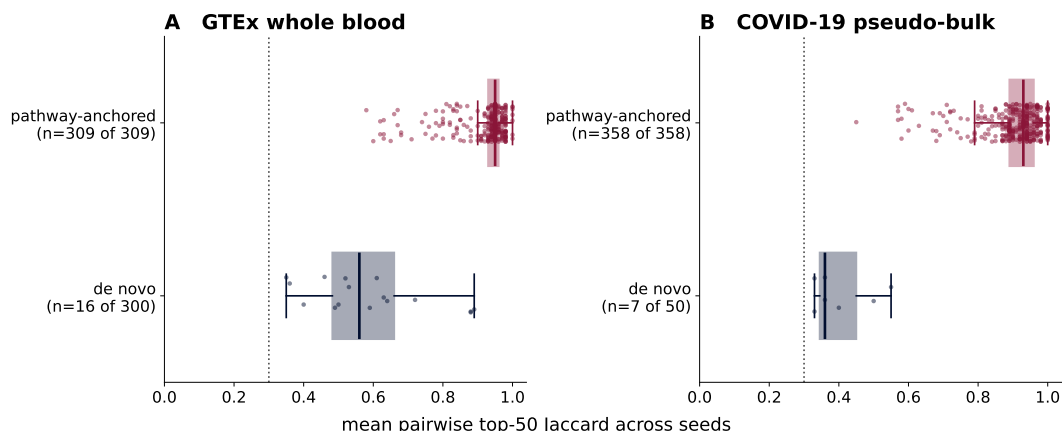


Fig. E6. **Cross-seed stability of the interpreted gene programs.** Mean pairwise Jaccard between the top-50 gene sets of a program’s members across five random seeds, for **(A)** GTEx whole blood (309 pathway-anchored + 300 free factors), and **(B)** COVID-19 pseudo-bulk (358 pathway-anchored + 50 free factors), both in combined mode. Boxes show the median and interquartile range with whiskers to $1.5 \times \text{IQR}$; points are individual programs. The dotted line marks the selection threshold ($J \geq 0.30$, together with presence in ≥ 3 of 5 seeds). Axis labels give the number of programs retained out of the number available in each class.

cohort with a larger free budget and read the programs from those fits. Table E1 gives the exact settings. Note that predictive results are never reported from these runs, and program-level results are never reported from the tuned predictive runs of Table D2.

The two program classes require different matching rules. A pathway factor is basically identified by its name, which is constant across seeds, and only the ranking of genes within the mask can vary. A *de novo* program has no fixed identity and hence, those programs are grouped into cross-seed clusters, where each factor joins an existing cluster with the highest mean Jaccard to its members, provided that the mean is at least 0.30, and otherwise seeds a new cluster. A program cluster is then called reproducible when it is present in at least 3 of the five initialization seeds (Fig.E6).

For each reproducible program we report a consensus top-50 list, ranking genes by their mean $\hat{s}_{g\ell}$ across seeds in which the program appears. Ensembl identifiers that could not be mapped to a gene symbol are reported as identifiers.

All pathway-masked programs pass the filter in both cohorts. This is expected rather than informative: because masked entries are held at $m_{g\ell} = 0$ and excluded from all updates (§B), a masked program’s gene support cannot change between seeds, and the Jaccard statistic measures only the stability of the within-mask ranking. The reproducibility result that carries information is therefore the *de novo* one, and it is far more selective.

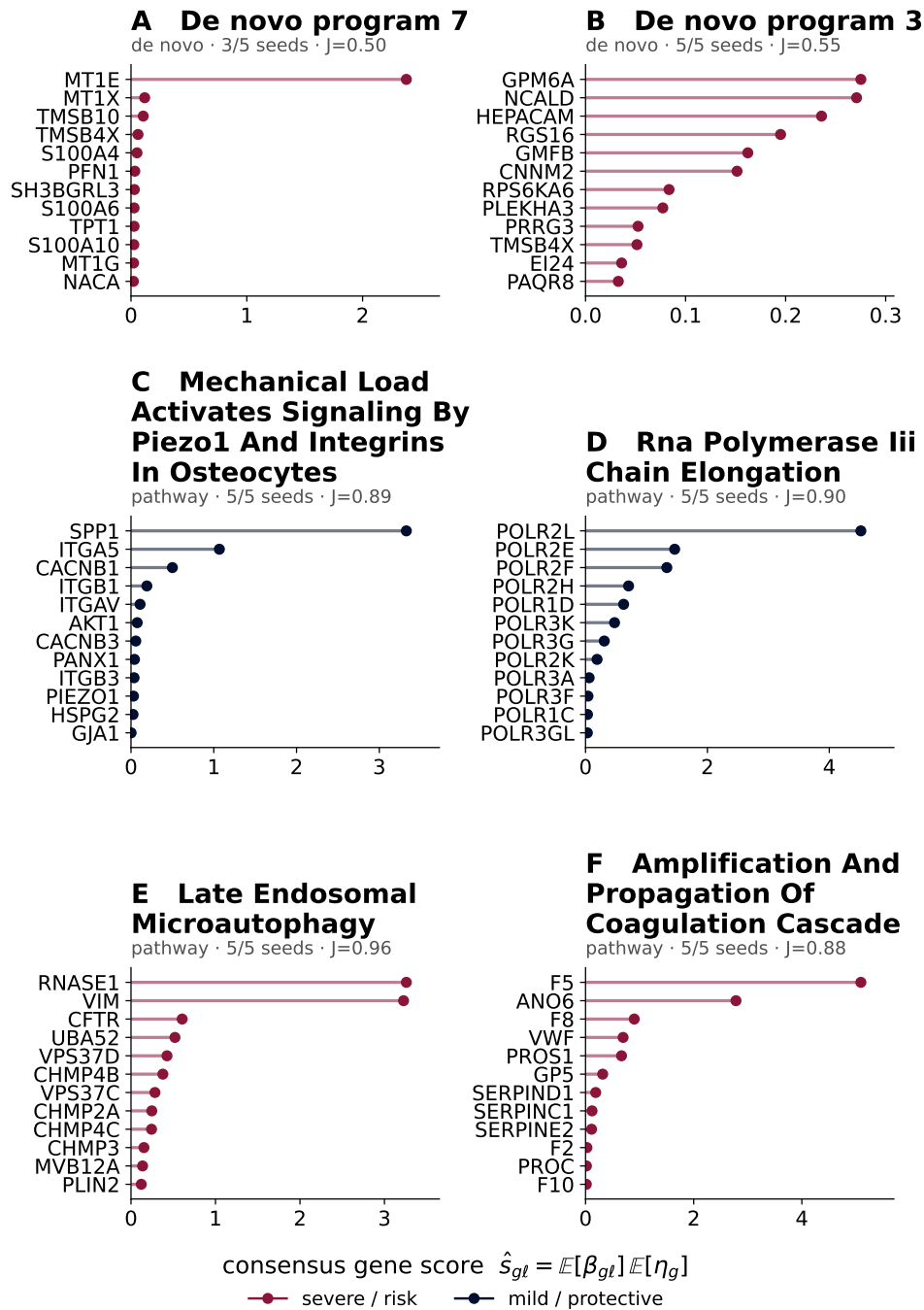


Fig. E7. **Reproducible COVID-19 gene programs and their leading genes.** The six programs with the largest effect-scaled importance $I_\ell = |\bar{v}_{k\ell}| \text{sd}(\bar{\theta}_\ell)$ among those passing the reproducibility filter(Appendix §E.3), from the pseudo-bulk combined fit with the Reactome immune mask. Each panel shows the top 15 genes of the consensus signature, ranked by the mean score $\hat{s}_{gl}$ across the seeds in which the program recurs. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards severe disease, navy for programs weighted towards mild disease.

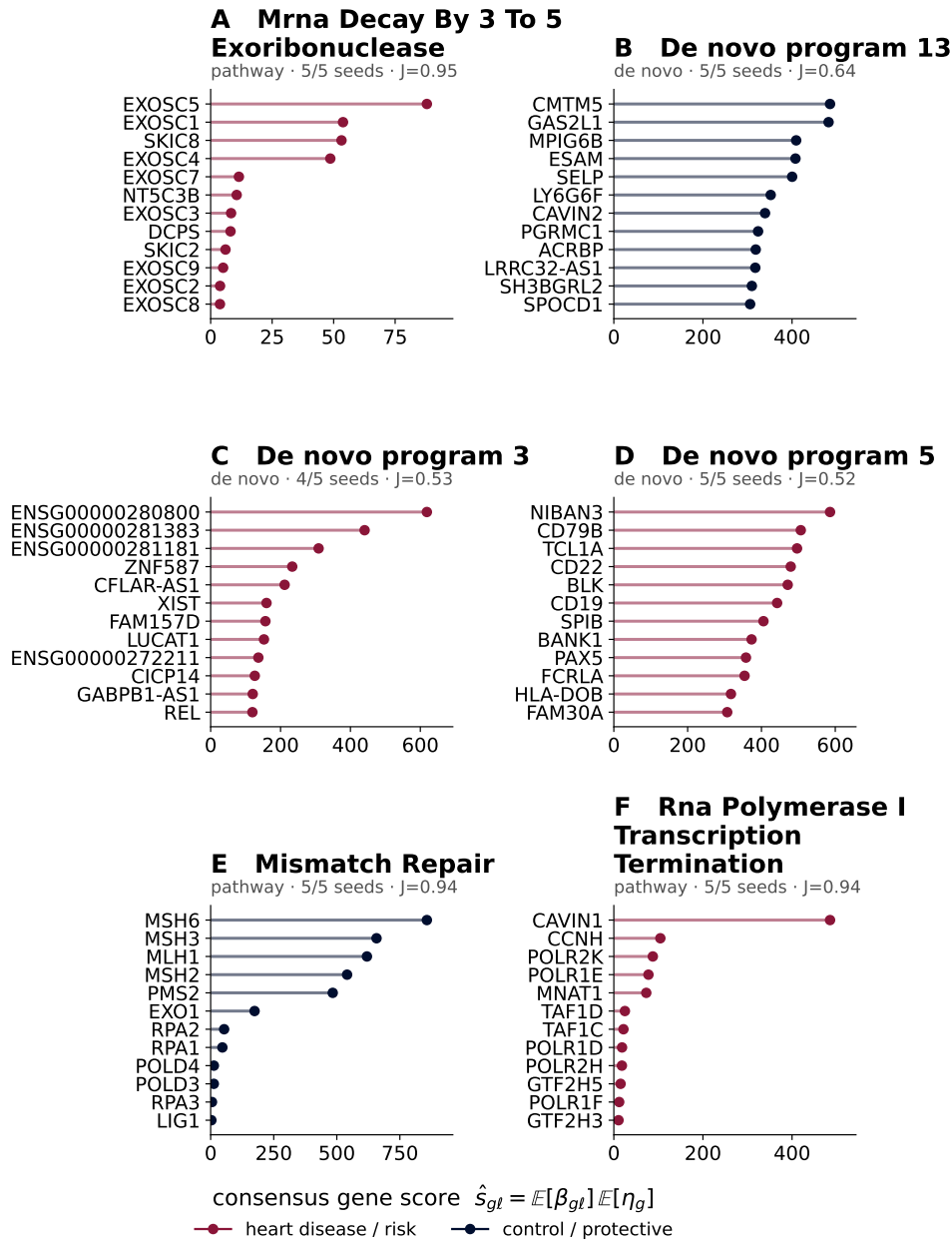


Fig. E8. **Reproducible GTEx heart disease gene programs and their leading genes.** The six programs with the largest effect-scaled importance I_ℓ among those passing the reproducibility filter(Appendix §E.3), from the GTEx heart disease combined fit with the Reactome immune mask. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards heart disease cases, navy for programs weighted towards controls.

Table E2. Disease-related genes within disease-associated gene programs.

Program	Dataset	Interpretation	Genes
GP7	COVID-19	Cell proliferation	<i>MKI67, TOP2A, CKS2, ZWINT, CENPF, CEP55, CCNB2, CDKN3, NCAPH, NUSAP1, PKMYT1</i>
GP7	COVID-19	Monocyte and macrophage markers	<i>MT1X, MT1G, S100A4, S100A6, LGALS1, COTL1, SRGN, HMG2</i>
GP7, GP3	COVID-19	Immune-regulatory metallothionein proteins	<i>MT1A, MT1E, MT1F, MT1G, MT1H, MT1M, MT1X</i>
GP7, GP3	COVID-19	Proteins that require binding to zinc or other metals to function	<i>ADNP, ARIH1, CDCA7, MAST4, NCALD, PAQR8, PKMYT1, PPM1N, PSPH, RPS6KA6, S100A4, S100A6, TOP2A, TRIM13, ZBTB20</i>
GP13	GTEEx	Cell adhesion	<i>CLDN5, EGFL7, MMRN1, SELP, TGFB11</i>
GP3	GTEEx	Megakaryocyte or platelet markers	<i>CAVIN2, CMTM5, GP6, PEAR1, SELP</i>
GP5	GTEEx	B cell markers	<i>BANK1, BLK, CD19, CD22, CD72, CD79B, FCRL1, FCRL2, FCRL5, FCRLA, HLA-DOB, IGHD, NIBAN3, PAX5, PLPPR5, TCL1A, VPREB3</i>

Appendix F. Runtime Analysis

We benchmarked wall-clock runtime across all four evaluation settings against the factorization baselines Spectra and scHPF and the non-factorization baselines (Figure F1; five seeds per method). In all experiments DRGP is the fastest factorization method, with median of 17 min on GTEx whole blood. On the bulk and single-cell simulations it is 4–5 $\times$ faster than scHPF and, on the single-cell simulation, more than an order of magnitude faster than Spectra (3 vs. 92 min). These gains follow from DRGP’s GPU implementation: the coordinate-ascent updates are expressed as fused, just-in-time compiled array operations over the sparse count matrix, so runtime scales with the number of nonzero entries rather than the dense cell by gene product. The advantage is most consequential at single-cell scale: on the full COVID-19 cohort (148 patients, $\sim$ 830k cells) DRGP factorizes the data on a single GPU, whereas scHPF exhausts available memory beyond 30 patients on a 500GB node, and Spectra required a 900GB node to complete the factorization on the full cohort, whereas DRGP required 4.7GB of memory on a single GPU.

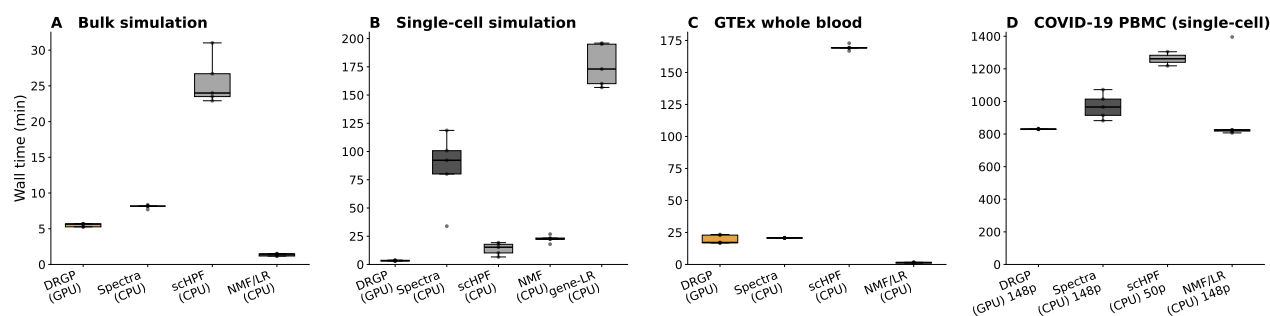


Fig. F1. Wall-clock runtime of DRGP (combined mode) against the baseline methods Runtime minutes on **A)** bulk RNA-seq simulations, **B)** single-cell simulations, **C)** GTEx whole blood bulk RNA-seq, and **D)** COVID-19 single-cell RNA-seq. Boxes show the median and interquartile range over five random seeds, whiskers extend to 1.5 $\times$ IQR, and points are the individual seeds. DRGP runs on a single GPU; scHPF, NMF, and the logistic-regression baselines run on CPU. on GTEx, Spectra was run on the 10,000 most-expressed genes while DRGP and the remaining baselines used the full 27,693-gene axis, and on COVID-19 the baselines used 13,770 genes. On COVID-19 PBMC each method is shown at the largest patient cohort it can process without exhausting memory: DRGP and the logistic/NMF baselines complete the full 148-patient cohort ($\sim$ 830k cells), whereas scHPF exhausts memory beyond 30 patients. Simulation and GTEx values are directly comparable across methods.

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